

Loops (Lecture 07)

Ques: Print all even numbers from 1 to 100

Summary Notes

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HW: Print all odd numbers divisible by 3
from 1 to 100

3 9 15 21 . . 99

Ques: Print the table of 19

19 38 57 76 95 114 133 152 171 190

HW : Print the table of 'n'.

Ques: Print numbers from 'n' to 1.

```
int n;  
cin >> n;
```

$n = 7$

Output - 7 6 5 4 3 2 1

$i \rightarrow 7 \text{ to } 1$

$i \geq 1$

$i = i - 1$ or $i--$

Ques: Display this AP - 2,5,8,11.. upto 'n' terms

$$a_n = a + (n-1)d$$

2, 5, 8, 11, 14, ...

$$a = 2$$

$$d = 3$$

n

```
for(i = a, i <= a_n, i += d) {  
    cout << i << " ";  
}
```

$$a_n = 2 + (n-1) \cdot 3 = 3 \cdot n - 1$$

Ques: Display this GP - 1,2,4,8.. upto 'n' terms

1 2 4 8 16 . . .

$$a_n = ar^{n-1} \rightarrow X$$

HW: Print this series - 99,95,91,87,.. upto all terms which are positive

$$a = 99$$

$$d = -4$$

$$a_n = ?$$

$$n = ?$$

Ques: Print all alphabets with their corresponding ASCII values.

A 65

B 66

C 67

D .

E .

F .

. .

. .

Z 90

HW: Take 'n' as input from user and print the following sequence..

1

n

2

n-1

3

n-2

...

Break & continue

```
for(int i=1; i<=10; i++){  
    if(i==5) break;  
    cout<<i<<" ";  
}
```

→ loop beech me
hi tod diya

Output

1 2 3 4

5
4
3
2
1
i

Break & continue



Skips an iteration

Break & continue

60 ke factors

1 2 3 4 5 6 $\sqrt{60}$ 10 12 15 20 30 60

if 'i' is a factor of n

then $\frac{n}{i}$ is also a factor of n

Ques: WAP to check if a given number is prime or not.

$i = 2$ to $n-1$ tak loop chala

if($n \% i == 0$) composite

Ques: Print all even numbers from 1 to 100